Magnetic Fields

Exam code: 0625 & 0972

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Magnetic fields

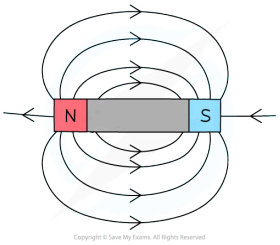
- A magnetic field is defined as:

A region in which a magnetic pole experiences a force

Magnetic field around a bar magnet

- The magnetic field is **strongest at the poles**
  - Therefore, the magnetic field lines are **closer** together at the ends of the magnets
- The magnetic field becomes **weaker as the distance from the magnet increases**
  - Therefore, the magnetic field lines get **further apart**

Magnetic field around a bar magnet



The magnetic field around a bar magnet

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

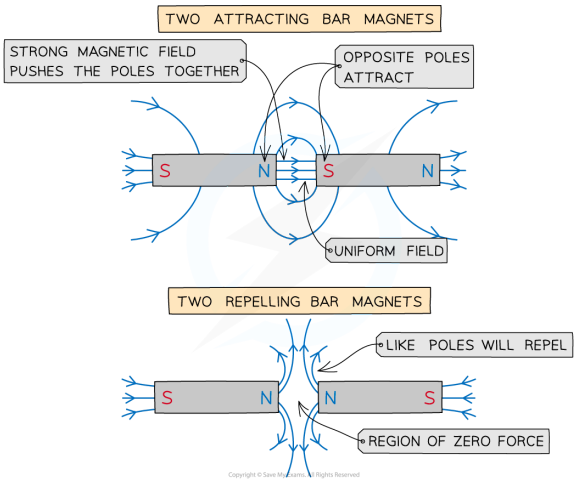
the field line
    - The direction of the field line shows the **direction** that the magnetic **force** would act
    - The field lines always go **from** a **north** pole **to** a **south** pole

Magnetic field strength

Extended tier only

- Magnetic forces are due to interactions between magnetic fields
    - Two bar magnets can repel or attract
- This is shown by the magnetic field lines between the two magnets:

Attractive and repulsive magnetic fields



Magnetic field lines for attracting and repelling bar magnets

- The **strength** of the magnetic field is shown by the **spacing** of the magnetic field lines
    - If the magnetic field lines are **close together** then the magnetic field will be **strong**
    - If the magnetic field lines are **far apart** then the magnetic field will be **weak**

Examiner Tips and Tricks

If you are asked to draw the magnetic field around a bar magnet remember to indicate both the **direction** of the magnetic field and the **strength** of the magnetic field. You can do this by:

the distance from the magnet increases

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